Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/04/2005
PI Name:	Palmieri, Maria Ph.D.		
Project Title:	The Role of the Actin Cytoskeleton in Gravitropic Signal Transduction in Arabidopsis thaliana		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	palmiem@muohio.edu	**Fax:**	FY 513-529-4243
PI Organization Type:	UNIVERSITY	**Phone:**	513-529-4200
Organization Name:	Miami University		
PI Address 1:	Department of Botany		
PI Address 2:	316 Pearson Hall		
PI Web Page:			
City:	Oxford	**State:**	OH
Zip Code:	45056	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
Start Date:	07/01/2004	**End Date:**	06/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The overarching goal of this project is to determine the role of the actin cytoskeleton in signal transduction mechanisms of plant gravitropism. This study characterizes plastid sedimentation kinetics in the statocytes (gravity sensing cells) of stem-like organs of Arabidopsis through analyses of (1) plastid movement in the statocytes of wild-type plants, (2) the effects of F-actin depletion on plastid sedimentation, (3) plastid movement in mutants with large plastids, and (4) videomicroscopy of GFP-labeled amyloplasts. The objective of the proposed research is to characterize plastid sedimentation kinetics in statocytes of Arabidopsis thaliana in order to elucidate the roles of amyloplasts and the F-actin cytoskeleton in shoot gravitropism. These goals are important because signal transduction is the least understood phase of plant gravitropism. We test the actin-tether and tensegrity hypotheses for gravity perception through pharmacological alteration of F-actin and the actomyosin system in wild-type and gravitropism mutants of Arabidopsis. Also, utilization of advanced visualization techniques (i.e., GFP constructs) provides us with an excellent opportunity to significantly increase our understanding of the role the cytoskeleton plays in gravitropic signal transduction. Results from this project		

	will promote increased understanding of the mechanisms of gravitropism, which, in the long-term, is essential when addressing the unique challenges of growing plants in microgravity. This last objective will be an important part of the exploration agenda because plants will be part of a bioregenerative life-support system on Mars.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research provides insights into fundamental signal transduction mechanisms in plant cells. Successful completion of this research will expand current knowledge as it pertains to cell/molecular biology and gravitational ecology, which are two specific objectives in the area of space biology. Increased understanding of plant gravitropism also is essential when addressing the unique challenges of growing plants in microgravity, which will be important on long-term spaceflight missions (e.g. Mars). Thus, this research aids the current exploration agenda at NASA.
Task Progress:	Over the past year, efforts have been focused on completion of the fixed-tissues phase of this project. This detailed study included determination of amyloplast positions at various intervals following seedling reorientation in endodermal cells of Arabidopsis hypocotyls, both before and after pharmacological disruption of the F-actin cytoskeleton. Data was obtained by fixing the hypocotyls at various time intervals following reorientation, embedding the specimens in resin, obtaining semithin sections and digital images of the hypocotyls, and then analyzing amyloplast positions in the endodermal cells using image analysis software. As opposed to the aqueous chemical fixations conventionally used for this type of study, our analysis was accomplished using a technically demanding cryofixation protocol that produces extremely rapid immobilization of plastids. We used this method because we found that amyloplasts in Arabidopsis hypocotyls settle much too quickly for aqueous chemical solutions to penetrate and fix the tissues before significant amyloplast movement occurs, and so rapid freezing was necessary to adequately arrest amyloplast movement. Another innovation was the development of custom embedding molds for use with LR White, the epoxy resin we used for our specimens. These molds are inexpensive and versatile, and provide a chamber from which oxygen can be excluded and in which the specimens can be flat-embedded. In total, the fixed-tissues portion of the project has resulted in 6 presentations, 6 published abstracts, 4 awards, 1 book chapter, and 2 manuscripts submitted to peer-reviewed journals. Our results provide the first analysis of amyloplast sedimentation kinetics in Arabidopsis hypocotyls. Results demonstrate that F-actin disruption in Arabidopsis hypocotyls results in amyloplast clustering and loss of mobility. This is in contrast to Arabidopsis roots, where cytoskeletal disruption enhances amyloplast movement. The disparate results between these studies indicate that 1) the gravitropic mechanisms differ between the roots and hypocotyls of Arabidopsis, and 2) an intact F-actin cytoskeleton is not necessary for gravity perception in these organs. However, results also suggest that F-actin is crucial for a normal gravitropic response, possibly functioning as a feedback mechanism to modulate gravitropic curvature and/or to terminate curvature. The next phase of this project delves deeper into the interactions between amyloplasts and F-actin, and how these interactions affect gravitropism. According to all present gravitropism models, it is intuitive that increased amyloplast movement might result in an enhanced gravity response. However, no current model satisfactorily explains why the inhibition of amyloplast movement concomitant with cytoskeletal disruption actually results in an enhanced graviresponse. We have modified and rotated a confocal microscope such that it possesses a rotatable vertical stage. Preliminary experiments are underway to analyze amyloplast movement in endodermal cells of Arabidopsis hypocotyls in vivo using different pharmacological agents. The goal of this portion of the project is to unequivocally establish the relationship between F-actin and amyloplast movement.
Bibliography Type:	Description: (Last Updated: 06/30/2006)
Abstracts for Journals and Proceedings	Palmieri, M.; Kiss, J.Z. "The Role of the F-Actin Cytoskeleton in Signal Transduction Mechanisms of Gravitropism in Arabidopsis Hypocotyls. " Annual Meeting of the American Society of Plant Biologists, Lake Buena Vista, Florida. Abstract Supplement of the Annual Meeting of the American Society of Plant Biologists, July 2004: p 109 , Jul-2004
Abstracts for Journals and Proceedings	Kumar, N.S.; Palmieri, M.; Kiss, J.Z. "The Rate of Plastid Sedimentation in the Hypocotyls of the arg1 (Altered Response to Gravity) Mutant of Arabidopsis." American Society for Gravitational and Space Biology annual meeting, New York, NY, November 2004. Gravitational and Space Biology Bulletin 2004 Nov;18(1): 13. , Nov-2004
Abstracts for Journals and Proceedings	Palmieri, M.; Kiss, J.Z. "Studies of Plastid Sedimentation Kinetics in Arabidopsis Hypocotyls Using a Cryofixation Method." American Society for Gravitational and Space Biology annual meeting, New York, NY, November 2004. Gravitational and Space Biology Bulletin 2004 Nov;18(1): 13. , Nov-2004
Articles in Peer-reviewed Journals	Palmieri, M.; Kiss, J.Z. "Disruption of the F-Actin Cytoskeleton Limits Statolith Movement in Arabidopsis Hypocotyls." submitted to: Journal of Experimental Botany, 14 April 2005. , Apr-2005
Articles in Peer-reviewed Journals	Palmieri, M.; Kiss, J.Z. "A Novel LR White Flat-Embedding Technique to Study Cryofixed Plant Cells." Microscopy Research and Technique: in preparation, April 2005. , Apr-2005
Awards	Palmieri, M. "Graduate Student Achievement Award. May 2004." May-2004
Books/Book Chapters	Palmieri, M.; Kiss, J.Z. "The Role of Plastids in Gravitropism." in "Advances in Photosynthesis and Respiration." Ed. Wise, R.R.; Hoober, J.K. Dordrecht: Kluwer Academic Publishers, in press 2005., Apr-2005